Smart Electricity Meter System

Project Portfolio

Smart Electricity Meter Using Arduino

ITC Project Portfolio

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1. Introduction

With the rapid growth of technology and the increasing demand for electricity management systems, smart electricity meters have become an important part of modern smart homes and industries. Traditional electricity meters require manual monitoring and provide limited information about electricity usage.

This project introduces a Smart Electricity Meter System using Arduino technology. The system is designed to monitor electricity consumption in real time, calculate power usage, and display the results through an LCD screen.

The project aims to provide a low-cost and efficient solution for monitoring electricity usage while improving user awareness about power consumption.

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2. Project Overview

The Smart Electricity Meter System is an embedded system project that measures and displays electrical power consumption using Arduino components. The system continuously monitors the current and voltage values and calculates the amount of consumed electrical energy.

The measured values are displayed on an LCD screen, allowing users to monitor their electricity consumption instantly.

This project can be used in homes, laboratories, and educational environments as a demonstration of smart energy monitoring systems.

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3. Problem Statement

Traditional electricity meters have several limitations, including:

- Manual meter reading.
- Lack of real-time monitoring.
- Difficulty tracking daily electricity usage.
- Limited user awareness about energy consumption.

- Inefficient electricity management.

This project addresses these problems by developing a smart and automated monitoring system.

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4. Project Objectives

The main objectives of the project are:

- To measure electrical power consumption.
- To display real-time electricity usage.
- To calculate total consumed energy.
- To provide a simple and affordable monitoring solution.
- To improve energy management awareness.
- To apply embedded systems concepts in real-world applications.

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5. Features of the System

The system provides several useful features:

- Real-time electricity monitoring.
- LCD display output.
- Power usage calculation.
- Low-cost implementation.
- Easy installation and operation.
- Compact and portable design.
- Educational demonstration for embedded systems.

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6. System Components

The project consists of the following main components:

Component	Description
Arduino UNO	Main microcontroller used for processing
LCD 16x2 Display	Displays system information
Current Sensor	Measures current consumption
Voltage Sensor	Measures voltage values

Component	Description
Breadboard	Used for prototyping
Jumper Wires	Connections between components
Power Supply	Provides power to the system

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7. Hardware Requirements

The hardware used in this project includes:

- Arduino UNO Board
- ACS712 Current Sensor
- Voltage Sensor Module
- LCD 16x2 Display
- Potentiometer
- Breadboard
- Connecting Wires
- USB Cable
- External Power Source

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8. Software Requirements

The software tools required are:

- Arduino IDE
- Proteus Simulation Software (Optional)
- Embedded C Programming Language

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9. System Architecture

The system architecture is divided into the following stages:

1. Input Stage
    - Current and voltage values are collected using sensors.
2. Processing Stage
    - Arduino processes the sensor readings.
3. Output Stage

    - LCD displays the electricity consumption values.
4. Calculation Stage
    - Energy consumption and billing calculations are performed.

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10. Circuit Design

The circuit design connects the sensors and LCD module to the Arduino board.

Main Connections:

- LCD RS → Arduino Pin 12
- LCD E → Arduino Pin 11
- LCD D4-D7 → Arduino Pins 5,4,3,2
- Current Sensor Output → Analog Pin A0
- Voltage Sensor Output → Analog Pin A1

The Arduino receives sensor data and processes it continuously.

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11. Working Principle

The system operates according to the following steps:

1. Sensors detect voltage and current.
2. Arduino reads the analog sensor values.
3. The program calculates power consumption.
4. Results are displayed on the LCD.
5. The system updates readings continuously.

The formula used:

Power (W) = Voltage × Current

Energy Consumption (kWh) is calculated over time.

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12. Arduino Code Explanation

The Arduino program performs the following tasks:

- Reads analog sensor values.

- Converts sensor readings into electrical values.
- Calculates power and energy consumption.
- Displays results on the LCD.
- Updates values in real time.

Example Code Snippet

```
float voltage = analogRead(A1);
float current = analogRead(A0);
float power = voltage * current;

lcd.setCursor(0,0);
lcd.print("Power: ");
lcd.print(power);
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13. LCD Display Functions

The LCD screen displays:

- Voltage value
- Current value
- Power consumption
- Total energy usage
- Estimated electricity bill

The display updates every few seconds to provide live monitoring.

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14. Power Consumption Calculation

The system calculates electricity usage using mathematical formulas.

Formula:

Energy (kWh) = Power × Time

Where:

- Power is measured in kilowatts.
- Time is measured in hours.

The system continuously updates the total consumed energy.

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15. Billing System

The project can estimate the electricity bill according to energy consumption.

Example Formula:

Bill = Energy Consumption × Cost Per kWh

This allows users to estimate their electricity expenses.

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16. Advantages of the Project

The Smart Electricity Meter offers many advantages:

- Real-time monitoring.
- Improved energy awareness.
- Cost-effective implementation.
- Easy maintenance.
- Portable and compact design.
- Educational value for students.

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17. Challenges Faced

During project development, several challenges were encountered:

- Sensor calibration issues.
- Reading fluctuations.
- LCD display synchronization.
- Circuit connection errors.
- Power stability problems.

These challenges were solved through testing and debugging.

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18. Future Improvements

The project can be improved in the future by adding:

- Wi-Fi connectivity.
- Mobile application integration.
- Cloud data storage.
- Automatic electricity cut-off system.
- Smart home integration.
- Online monitoring dashboard.

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19. Applications

This system can be used in:

- Smart homes
- Schools and universities
- Industrial monitoring systems
- Energy management systems
- Educational laboratories

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20. Conclusion

The Smart Electricity Meter System successfully demonstrates the use of embedded systems in monitoring electrical energy consumption.

The project provides an affordable and efficient solution for real-time electricity monitoring using Arduino technology. It also improves user awareness regarding electricity usage and energy management.

This project helped develop practical skills in embedded systems, programming, circuit design, and problem solving.

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21. References

1. Arduino Official Documentation
2. Embedded Systems Fundamentals
3. Smart Meter Research Papers
4. Arduino IDE Documentation
5. Electronics and Sensor Datasheets
